

DEEPAK PATKAR

PhD, Theoretical and Computational Chemistry

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Experience	Post-doctoral Fellow Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences Research Topics: Copper-Peptide mediated Liquid-Liquid Phase Separation ML-Predictive Design of MOFs for SO ₂ , H ₂ S, and NH ₃ Capture Skills: GROMACS AmberTools LAMMPS RASPA VASP Materials Studio Quantum ESPRESSO CP2K	2025-Present
	Subject Matter Expert at Turing AI Responsibilities: Image-based LLM data curation, annotation, and quality review. Skills: Data Curation Data Annotation Quality Assurance Image Analysis Content Review	2025-
	Senior Associate at Innodata India Private Limited Role: Subject Matter Expert Responsibilities: Image-based LLM training data curation Skills: Data Curation Data Annotation Quality Assurance Image Analysis Content Review	2024-2025
Academics	Doctor of Philosophy, Chemistry Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India Thesis Supervisor: Dr. Milind Madhusudan Deshmukh [link] Thesis: <i>Quantum Chemical Studies on the Mixed Molecular Clusters: Quantitative Estimates of Hydrogen Bond Strength and Cooperativity.</i> [link] Skills: Gaussian AIMAll DAmQT Multiwfn GaussView ChemDraw VMD Avogadro Data Analysis ML Techniques Cheminformatics MTA Catalysis NCI MESP QTAIM DFT Theoretical Chemistry	2019-2024
	Master of Science, Chemistry Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India Project Title: Application of HPLC and Gas Chromatography for Chemical Analysis. Project Supervisor: Dr. Abhilasha Durgavanshi [link] Skills: Linux Python FORTRAN Bash Scripting HPLC GC-MS Analytical Chemistry Natural Product Chemistry Quantum Chemistry Thermodynamics Spectroscopy Material Chemistry	2017-2019
	Bachelor of Science, Triple major in Physics, Chemistry, Criminology and Forensic Science Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India Built foundational understanding of principles, techniques across diverse disciplines. General Classical Physics General Quantum Physics Physical Chemistry Organic Chemistry Inorganic Chemistry Analytical Chemistry Forensic Science Toxicology Criminology Crime Scene Investigation	2014-2017
Research Interest	Theoretical Chemistry Density Functional Theory Noncovalent Interactions Cooperative Effects Transition Metal Catalysis Molecular Dynamics Simulations Protein Simulations Machine Learning Cheminformatics Bioinformatics	
Publications	Total Publication = 8 , h-index = 6 , i10-index = 5 , Citations = 143 Google Scholar ID: DxGUFx0AAAAJ	
	- Ismail, T. M.; Patkar, D.; Sajith, P. K.; Deshmukh, M. M. Hydrogen bond strengths in microhydrated clusters of HNO and HONO: Energetic insights via molecular tailoring approach <i>New J. Chem.</i> 2024, 48, 11591–11605. [link] - Ismail, T. M.; Patkar, D.; Sajith, P. K.; Deshmukh, M. M. Interplay of Hydrogen, Pnictogen and Chalcogen Bonding in X(H₂O)_n=1-5 (X = NO, NO⁺, and NO⁻) Complexes: Energetics Insights via Molecular Tailoring Approach <i>J. Phys. Chem. A</i> 2023, 127, 10360–10374. [link] Patkar, D.; Chopra, D.; Deshmukh, M. M. Characterization of non-covalent contacts in mono- and di-halo substituted acetaldehydes: probing the substitution effects of electron donating and withdrawing groups <i>Phys. Chem. Chem. Phys.</i> 2023, 25, 2946. [link] Patkar, D.; Ahirwar, M. B.; Deshmukh, M. M. A Tug of War between the Self- and Cross-associating Hydrogen Bonds in Neutral Ammonia-Water Clusters: Energetic Insights by Molecular Tailoring Approach. <i>ChemPhysChem</i> 2022, 23, e202200476. [link] Patkar, D.; Ahirwar, M. B.; Deshmukh, M. M. Energetic Ordering of Hydrogen Bond Strengths in Methanol-Water Clusters: Insights via Molecular Tailoring Approach <i>ChemPhysChem</i> 2022, 23, e202200143. [link] Patkar, D.; Ahirwar, M. B.; Shrivastava, S. P.; Deshmukh, M. M. Assessment of Hydrogen Bond Strengths and Cooperativity in Self- and Cross-Associating Cyclic (HF)_m(H₂O)_n (m + n = 2 to 8) Clusters. <i>New J. Chem.</i> 2022, 46, 2368–2379. [link]	

	7. Patkar, D.; Ahirwar, M. B.; Gadre, S. R.; Deshmukh, M. M. Unusually Large Hydrogen-Bond Cooperativity in Hydrogen Fluoride Clusters, (HF)_n, n = 3 to 8, Revealed by the Molecular Tailoring Approach. <i>J. Phys. Chem. A</i> 2021, 125, 8836–8845. [link] 8. Ahirwar, M. B.; Patkar, D.; Yadav, I.; Deshmukh, M. M. Appraisal of Individual Hydrogen Bond Strengths and Cooperativity in Ammonia Clusters via Molecular Tailoring Approach. <i>Phys. Chem. Chem. Phys.</i> 2021, 23, 17224–17231. [link] 9. Patkar, D. Uncovering the Individual Hydrogen Bond Strengths and Cooperativity in Pure (H₂S)_n and Mixed (H₂O)_m(H₂S)_n (m + n = 2 to 8) Clusters. <i>ChemRxiv</i> 2025. [link] [Pre-print] 10. Patkar, D. From Weak to Functional: Network Cooperativity and Anti-Cooperativity in Cyclic Neutral (HF)_m(NH₃)_n Hydrogen Bonded Clusters, <i>J. Phys. Chem. A</i> 2026, xxx,xxxx–xxxx. [Submitted]
Selected Talks	1. National Seminar on “Recent Advances in Chemical Sciences”, Department of Chemistry, Dr. Harisingh Gour Vishwavidyalaya Sagar, March, 2023. [link] Title: Competition between Self- and Cross-associating Hydrogen Bonds in Neutral Ammonia-Water Clusters: Energetic Insights by Molecular Tailoring Approach. [link] 2. DAE Symposium on “Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September, 2022. [link] Title: Energetic Ordering of Hydrogen Bond Strengths in Methanol-Water Clusters: Insights via Molecular Tailoring Approach. [link] 3. DAE Symposium on “Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September, 2021. [link] Title: Evaluation of individual Hydrogen Bond Strengths and Cooperativity in (HF)_m(H₂O)_n, (m + n) = 2 to 8 Clusters. [link] 4. Theoretical Chemistry Symposium, IISER-Kolkata, IACS Kolkata, December, 2021. [link] Title: Estimation of Individual Hydrogen Bond Strengths and Cooperativity in (HF)_m(H₂O)_n, (m + n) = 2 to 8 Clusters via Molecular Tailoring Approach. [link] 5. International Seminar, Dr. Harisingh Gour Vishwavidyalaya Sagar, January, 2021. Title: Theoretical Study of Hydrosilylation of Olefin Catalyzed by Cationic Calcium Hydride Catalyst. [link]
Selected Conferences	1. National Seminar on “Recent Advances in Chemical Sciences”, Department of Chemistry, Dr. Harisingh Gour Vishwavidyalaya Sagar, March 2023. [link] 2. DAE Symposium on “Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September 2022. [link] 3. Hand on Workshop “Molecular Tailoring Approach”, Department of Scientific Computing, Modelling and Simulation, Savitribai Phule Pune University, May 2022. [link] 4. Theoretical Chemistry Symposium, IISER-Kolkata, IACS Kolkata, Kalyani University, December, 2021. [link] 5. DAE Symposium on “Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September 2021. [link]
Other Skills	Molecular Modeling ChemCraft Spartan PyMOL ChimeraX VESTA OVITO RDKit PySCF LaTeX Gnuplot OriginPro Microsoft Excel Scientific Programming Data Analysis Scientific Visualization HPC Linux Quantum Computing Machine Learning Cheminformatics Bioinformatics Python Bash HTML FORTRAN C R
Academic Achievements	INSPIRE Scholarship, DST India (2014) Best Oral Presentation Award, DHSGSU (2023) Member, American Chemical Society (2021–Present)
Teaching Experience	Teaching Assistant, Dr. Harisingh Gour Vishwavidyalaya, Sagar, India Conducted undergraduate chemistry laboratory classes Developed computational chemistry lab assignments Guided students in scientific computing
References	1. Dr. Milind Deshmukh Ph.D. Supervisor Associate Professor, Department of Chemistry, University of Delhi, Delhi Mobile: +91 8989180813, Email: milind.deshmukh@gmail.com webpage: [link] 2. Prof. Deepak Chopra Crystallography and Crystal Chemistry Laboratory, Department of Chemistry, IISER Bhopal Office: +91-755-669-1311; Fax: +91 755 6692392, Email: dchopra@iiserb.ac.in webpage: [link] 3. Dr. Pookkottu K. Sajith Assistant Professor, Department of Chemistry, Farook College, Kozhikode, Kerala Mobile: +91 9349280803, Email: psajith@farookcollege.ac.in webpage: [link]